满分：100分

99分

此处未上传图片

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13.（6 分）（1） D

（2） ~~0.50 J~~ 0.49J　0.48 J

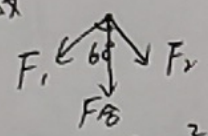
14.（4 分）（1） x_A，x_0，x_B ；

（2） $m_A x_0 = m_A x_A + m_B x_B$

15.（8 分）

解：∵等边三角形

∴ ~~R=~~ $r=\frac{\sqrt{3}}{3}R$



$F_1=F_2=G\frac{m^2}{R^2}$

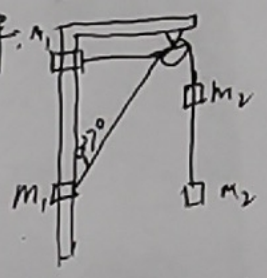
∴ $F_合=G\frac{\sqrt{3}m^2}{R^2}$

$F_合=m(\frac{2\pi}{T})^2 r$

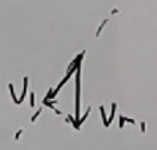
∴ $T=\sqrt{\frac{4\pi^2R^3}{3Gm}}$

16.（10 分）

解：



m_1速度



$V_1=V_{m_1}\cos37°=V_{m2}$

~~mg~~ 根据动能定理，m_1下降高度 $h=4d$，

m_1下降后与滑轮距离 $x=5d$

$m_1gh-m_2g(x-3d)=\frac{1}{2}m_1V_{m_1}^2+\frac{1}{2}m_2V_{m_2}^2$

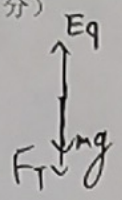
~~$V_{m2}=\frac{10}{10}\sqrt{214}$ m/s~~

$V_{m2}=\frac{8}{7}$ m/s

-1

17.（10 分）

解.(1)



$Eq = 8N$

$mg = 4N$

$\because Eq > mg$

$\therefore F_T$方向向下

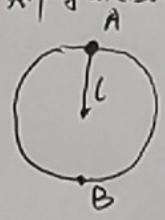
$\therefore F_T = Eq - mg = 4N$

(2) 将 Eq 与 mg 合并为等效场

$a_1 = \frac{(Eq - mg)}{m} = 10\ m/s^2$

方向竖直向上

则等效圆周运动最高处为B处



$ma_1 = m\frac{v_B^2}{l}$

$v_B = \sqrt{a_1 l} = 2\ m/s$

$\therefore \frac{1}{2}mv_A^2 = ma_1 \cdot 2l + \frac{1}{2}mv_B^2$

$\therefore v_A = v_0 = 2\sqrt{5}\ m/s$

18.（14 分）

解.(1) $E_{pA} = \mu mg \cdot \frac{L}{2} + \frac{1}{2}mv_1^2$

$v_1 = \sqrt{10}\ m/s$

(2) $mv_1 = mv_0 + mv_2$

$\frac{1}{2}mv_1^2 = \frac{1}{2}mv_0^2 + \frac{1}{2}mv_2^2$

$\therefore v_0 = 0,\ v_2 = v_1 = \sqrt{10}\ m/s$

$\frac{1}{2}mv_2^2 = \mu mg \cdot \frac{L}{2} + \frac{1}{2}mv_D^2$

$v_D^2 = 5\ m^2/s^2$

$\because$ 小恰好通过D点

$\therefore mg = m\frac{v_D^2}{R}$

$\therefore R = 0.5m$

(3) $\frac{1}{2}mv_D^2 + mg \cdot 2R = \frac{1}{2}mv_N^2$

$\therefore v_N = 5m/s$，方向向左

$ma_1 = \mu mg$

$a_1 = 5\ m/s^2$

$a_1 t = v_N - (-v) = 9\ m/s$

$\therefore t = 1.8s$

$\therefore a_1 t_1 = v_N$

$t_1 = 1s$

$x_{相} = vt_1 + \frac{v_N^2}{2a_1} = 6.5m$

$a_1 t_2 = v$

$t_2 = 0.8s$

$x'_{相} = vt_2 - \frac{1}{2}a_1 t_2^2 = 1.6m$

$\therefore Q = \mu mg(x_{相} + x'_{相}) = 8.1J$